Impact of Macroeconomic Indicators on Stock Market Performance (US Market)

Introduction

The stock market is influenced by various macroeconomic factors such as inflation, interest rates, unemployment, and money supply. This project aims to analyze the relationship between these macroeconomic indicators and stock market performance using statistical and econometric techniques.

The S&P 500 index is used as a proxy for overall market performance, while key macroeconomic variables are sourced from the Federal Reserve Economic Data (FRED) database.

Objective

The objective of this study is to examine how macroeconomic variables influence stock market movements and to identify which factors have the most significant impact.

Data Description

The data used in this project includes:

- S&P 500 Index (SP500) – Represents overall stock market performance
- Consumer Price Index (CPI) – Measures inflation
- Unemployment Rate – Indicates economic health
- Federal Funds Rate – Represents interest rate levels
- Money Supply (M2) – Reflects liquidity in the economy

The data was obtained from the FRED database and processed using R.

Methodology

The analysis was conducted using multiple linear regression to examine the relationship between macroeconomic variables and stock market performance.

Log transformation was applied to selected variables to stabilize variance and improve model accuracy.

The regression model is specified as:

SP500 = f(CPI, Unemployment, Interest Rate, Money Supply)

Several diagnostic tests were performed to validate the model, including tests for normality, multicollinearity, heteroskedasticity, autocorrelation, and stationarity.

Results and Interpretation

Regression Analysis

The regression model was used to analyze the impact of macroeconomic variables on stock market performance (S&P 500).

The results show:

- CPI (Inflation) has a positive and statistically significant relationship with the stock market ($p < 0.001$), indicating that increases in inflation are associated with higher stock market values.
- Interest Rate also shows a positive and significant impact ($p < 0.001$), suggesting that changes in interest rates influence market performance.
- Money Supply (M2) has a strong positive and highly significant effect ($p < 0.001$), indicating that higher liquidity in the economy supports stock market growth.
- Unemployment has a negative but statistically insignificant relationship, suggesting that its impact on the stock market is weak in this model.

Model Diagnostics

Normality Test (Jarque-Bera)

The Jarque-Bera test resulted in a **p-value < 0.05**, indicating that the residuals are **not normally distributed**.

Multicollinearity (VIF Test)

The VIF values for all variables are below 5:

- CPI ≈ 1.70
- Unemployment ≈ 2.30
- Interest Rate ≈ 1.77
- Money Supply ≈ 1.29

This indicates that **multicollinearity is not a concern** in the model.

Heteroskedasticity (Breusch-Pagan Test)

The Breusch-Pagan test shows a **p-value < 0.05**, indicating the presence of **heteroskedasticity** in the model.

Autocorrelation (Durbin-Watson Test)

The Durbin-Watson test shows a **very low statistic (~0.27)** with a **significant p-value**, indicating **strong positive autocorrelation** in the residuals.

Stationarity Test (ADF Test)

The Augmented Dickey-Fuller test results indicate:

- **SP500** → Stationary ($p < 0.05$)
- **CPI** → Stationary ($p < 0.05$)
- **Unemployment** → Not stationary ($p > 0.05$)
- **Interest Rate** → Not stationary ($p > 0.05$)
- **Money Supply** → Not stationary ($p > 0.05$)

This suggests that some variables may require further transformation for improved model accuracy.

Correlation Analysis

The correlation matrix shows:

- Strong positive correlation between **Money Supply and SP500 (~0.84)**
- Moderate positive correlation between **CPI and SP500 (~0.51)**
- Negative correlation between **Unemployment and SP500 (~ -0.47)**

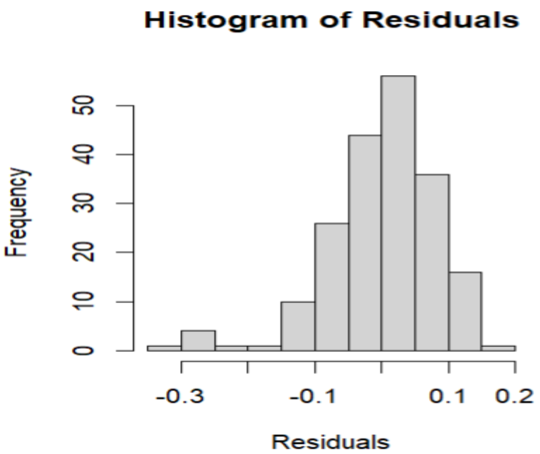
This supports the regression findings and indicates that liquidity and inflation are key drivers of market performance.

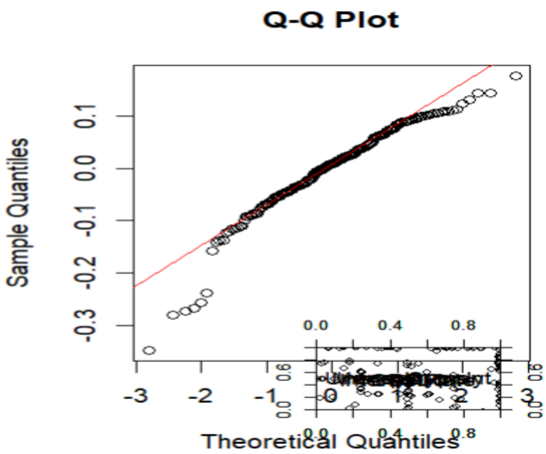
Residual Analysis

The histogram and Q-Q plot of residuals show that:

- Residuals are approximately centered around zero
- Some deviations from normality are observed
- The Q-Q plot shows slight departures from the straight line

This suggests minor violations of normality assumptions.





Conclusion

The analysis provides strong evidence that macroeconomic factors play a crucial role in shaping stock market performance. Among the variables studied, **money supply (M2)** and **inflation (CPI)** emerge as the most influential drivers, both exhibiting a significant positive relationship with the S&P 500 index. This suggests that increased liquidity and expansionary economic conditions tend to support higher market valuations.

Interest rates also demonstrate a statistically significant impact on the stock market. While traditionally higher interest rates are associated with reduced valuations due to increased borrowing costs, the results indicate that their relationship with market performance can be complex and influenced by broader economic conditions. **Unemployment**, on the other hand, shows a negative but statistically insignificant effect, indicating that its direct impact on stock market movements may be limited or indirect within this model.

The correlation analysis further reinforces these findings, particularly highlighting the strong relationship between money supply and market performance, emphasizing the importance of liquidity in financial markets.

However, the diagnostic tests reveal certain limitations in the model. The presence of **heteroskedasticity** and **autocorrelation** suggests that the model does not fully capture the underlying time-series dynamics of the data. Additionally, the stationarity tests indicate that some variables are non-stationary, which may affect the reliability of the regression results.

Despite these limitations, the model provides valuable insights into the relationship between macroeconomic variables and stock market behavior. It highlights that investors and analysts should consider macroeconomic indicators, particularly liquidity and inflation, when making investment decisions.

In conclusion, this study demonstrates that while macroeconomic factors significantly influence stock market performance, a more refined approach—such as incorporating time-series models or additional variables—could further enhance predictive accuracy and provide deeper insights into market dynamics.